

Dynamic Games with Strategic Interactions: Framework for Analysis.

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March 11, 2026

- To be realistic a dynamic model of even the simplest of markets has to be complex; too complex to admit analytic results with much applied content.
- This is the reason that the theory papers on dynamics work in a highly “stylized” environment; i.e. an environment that is designed more for the clarity it allows in developing intuitions than for its realism.
- So we give up on the goal of analytic elegance. Instead we provide a framework for obtaining quantitative responses for dynamic policy or descriptive analysis in a more realistic, user specified, environment.
- The goal is to enable us to analyze the equilibrium implications of the more complex institutional structures that we actually observe. We delay the questions of whether it is reasonable to assume equilibrium and if not how we get there.
- To do so we take realistic (typically estimated) primitives and a notion of equilibrium, compute equilibrium policies, and use these policies to generate quantitative responses to changes from a pre-specified initial condition.

Basic Model: (Ericson and Pakes, 1995, Restud).

- I begin with a full information equilibrium model¹. This requires agents to have extensive information. In particular at every "state" (see below) each agent needs to have correct perceptions of the probability distribution of what their competitors' actions will be and best respond to that.
- I view this as a rest point; no agent has an incentive to deviate and if an agent is not best responding it has an incentive to deviate.
- It may not be the best way to match a model to data on the responses to a change but its used to approximate where we are headed. Moreover there are ways to compute a new equilibria that embody learning rules that select out likely equilibria in multiple equilibria situations.

¹For a more formal treatment see Doraszelski and Pakes (2007 *Handbook of Industrial Organization*). Here I will ignore random entry and exit fees for simplicity. Proofs of existence require this (see Doraszelski and Satterthwaite 2010, *RAND*). However many examples do not, and still compute equilibria (or at least " ϵ -equilibria").

States.

- $i \in \mathcal{Z}^+$.
- s_i will be the number of firms with efficiency level i ,
- $s = [s_i; i \in \mathcal{Z}^+]$ is the “industry structure” (a counting measure; the number of firms at each different efficiency level).
- *Assumption.* Given investment decisions, the distribution of future values of the state of the system (of (i, s)) are independent of the choice of prices or quantities.
 - Simple case where changes in price (quantity) affect only current profits, which implies $\pi(i, s)$ can be calculated “off-line” (without needing to compute the value function) so it simplifies calculations.
 - This generalizes the single agent investment model Myrto taught.
 - Subsequent lectures will focus on extensions; l.b.d., durable goods, collusion, asymmetric information.

Differentiated Product Example to carry along.

Demand

$$U_{i,j} = v_j - p_j^* + \epsilon(i, j)$$

where v_j is the “quality” of the good and p_j^* is the price (for $j = 1, \dots, J$).

Consumer i chooses good j if and only if

$$\epsilon(i, j) - \epsilon(i, q) > [v_q - v_j] - [p_q^* - p_j^*]$$

$$\equiv [v_q - \zeta] - [p_q^* - p_0^*] - \{[v_j - \zeta] - [p_j^* - p_0^*]\}$$

where ζ is the quality of the outside alternative,

$$\equiv i_q - p_q - [i_j - p_j]$$

which implicitly defines “real prices” and “real quality” (i.e. both relative to the outside alternative).

Define

$$C(i_j; s, p) = \{\epsilon : j = \max_q U_{i,q}\} \Rightarrow \sigma(i_j; s, p) = \frac{\exp[i_j - p_j]}{[1 + \sum \exp[i_q - p_q]]}.$$

Profits, assuming a constant marginal cost of mc are then

$$[p_j - mc]M\sigma(i_j; s, p).$$

The Nash pricing equilibrium pricing vector satisfies²

$$-[p_j - mc]\sigma_j[1 - \sigma_j] + \sigma_j = 0, \Rightarrow \pi(i_j; s) = [p(i_j; s) - mc]M\sigma[i_j; s, p(s)].$$

- Note: The presence of the outside alternative here is what insures the industry evolves, even in equilibrium.

²We are implicitly assuming uniqueness. For this simple case it is true, see Caplin and Nalebuff *Econometrica* 1991.

Bellman Equation for Incumbent Behavior.

$$V(i, s) = \max\{\phi, \pi(i, s) + \sup_{(x \geq 0)} [-cx + \beta \sum V(i', s') pr(i', s' | x, i, s)]\}. \quad (1)$$

where

- $pr(i', s' | x, i, s) = pr(i' | i, x) pr(s' | i, x, s)$, the “capital accumulation games” (definition, e.g.s it doesn't work, auctions)
- $i_{t+1} - i_t \equiv \tau_{t+1}$
- $\tau_t \equiv \nu_t - \zeta_t$
- ν = firm's investment outcome.

$$\mathcal{P} = \{p_\nu(\cdot | x), x \in \mathcal{R}^+\}$$

is stochastically increasing in x , and $Pr\{\nu > 0 | x = 0\} = 0$.

- ζ = common industry shock. Density $\mu(\zeta)$.

- Let
 - $\hat{s}_i = s - e_i$, where $e(i)$ has a one in the i^{th} slot and zero elsewhere, so $\hat{s}_i$ = the states of the competitors of a firm at state i .
 - $q[\hat{s}'_i | i, s, \zeta]$ provide the firm's perceived probability of its competitors future states conditional on a particular value of ζ . $q[\cdot | i, s, \zeta]$ embodies the incumbent's beliefs about entry and exit. Without an additional equilibrium assumption, this just says firms have beliefs, they need not be "correct". We come back to cases where we allow for incorrect beliefs.

$$pr(i' = i^*, s' = s^* | x, i, s) = \sum_{\zeta} p(\nu = i^* - i - \zeta | x) q[\hat{s}'_i = s^* - e(i^*) | i, s, \zeta] \mu(\zeta).$$

- **Entry model.** Many ways to model entry. One simple example.
 - Entrant must pay an amount $x_e (> \beta\phi)$ to enter, & enters one period later at state $i^e \in \Omega^e \subset \mathcal{Z}^+$ with probability $p^e(\cdot)$.
 - Only enters if the EDV of net cash flow $> x_e$.
 - x_e either fixed or random. When random only the potential entrant knows its value (rivals know it is a draw from the distribution)

Dynamic Equilibrium: Characterization Results.

- Notion dates back to Star and Ho (1969). Framework for I.O. from EP (1995). Existence proof Doraszelski and Satterthwaite (2003).
- Equilibrium conditions
 - Every agent chooses optimal policies given its perceptions on likely future industry structures.
 - Those perceptions are consistent with the behavior of the agent's competitors.
- Characterization results.
 1. It is “computable”, i.e..
 - Regularity conditions insure there is never more than $\bar{n}$ firms active.
 - Only observe “ i ” on $\Omega = \{1, \dots, K\}$ (given the competitors states, there is a lower i at which the given firm exits, and an upper i at which the given firm stops investing.).

1 $\Rightarrow$ equilibria for $(i, s) \in \Omega \times S$, where

$$S \equiv \{s = [s_1, \dots, s_k] : \sum s_j \leq \bar{n} < \infty\}$$

so that the number of elements in S or $\#S \leq K^{\bar{n}}$.

2. Generates a Markov chain for industry structures [for $\{s_t\}$].

$$Pr[s_{t+1} = s' | s^t] = Pr[s_{t+1} = s' | s_t]$$

and if the game is infinite horizon it is homogeneous w.r.t. time

$$Pr[s_{t+1} = s' | s_t] \equiv Q[s' | s_t].$$

$Q(\cdot, \cdot)$ on $S \times S$ is the "transition kernel" [$Q_t(\cdot, \cdot)$ if finite horizon].

3. They provide conditions on the primitives that insure that any equilibrium $Q[\cdot | \cdot]$ is *ergodic*. [Picture].

Ergodicity

- The states in R are "recurrent", and are the only states that are hit repeatedly. Other states are "transient".
- This is the analogue of the invariant measure in anonymous sequential games. So counterfactuals no longer focus on a single "industry structure" but a stochastic process which moves between industry structure.
- There can be multiple possible recurrent classes from a given set of policies (Polya urn schemes & Brian Arthurs motor example), but our regularity conditions generally rule them out.
- R is frequently much smaller than S (and the divergence is greatest for large markets with many state variables).
- The states in R , how they cycle through R , and the transitions to R depend on the primitives of the underlying economic model..

Brute Force Computation .

- This uses a "backward" solution technique from Pakes and McGuire (1994, *Rand*, use its computational example). Analogous to what you did in single agent games.
- It solves for strategies at every point in a discrete state space ($\# S \times S$ is finite). Initially assume that we know S (i.e. Ω and $\bar{n}$) is known.
 - In memory. Estimates of the value function and policies associated with each $(i, s) \in \Omega \times S$.
 - Updating. *Synchronous*; i.e. it circles through the points in S in some fixed order and updates all estimates associated with every $s \in S$ at each iteration (updates at each (i, s) that has $s_i > 0$ & updates entry).
 - Convergence. The values and policies from successive iterations are the same. Implies equilibrium conditions (for both values and policies).

Updating 1: Rewrite Bellman Equation.

$$V(i, s) = \max_{\chi \in \{0,1\}} \{[1 - \chi]\phi + \chi\{\pi(i, s) - \sup_{x \geq 0} [-cx + \beta \sum_{\nu} w(\nu; i, s)p(\nu|x_1)]\},$$

where

$$w(\nu; i, s) \equiv \sum_{(\hat{s}'_i, \zeta)} V(i + \nu - \zeta, \hat{s}'_i + e(i + \nu - \zeta)|w) q[\hat{s}'_i | i, s, \zeta] \mu(\zeta), \quad \&$$

$$q[\hat{s}'_i = s_i^* | i, s, \zeta] \equiv Pr\{\hat{s}'_i = \hat{s}_i^* | i, s, \zeta, \text{equilibrium policies}\}$$

- $w(\nu; i, s)$ is the EDV conditional on the current year's investment resulting in a particular value of ν , and the current state being (i, s) (it integrates out over the possible outcomes of both the investment strategies of competitors (the $\hat{s}'_i$), and over the outside alternative (the ζ)).

Updating Rules.

- $w(\nu; i, s)$ is all the firm needs to know to make decisions (a "sufficient statistic" for decision making); knowing it we have a single agent maximization problem.
- With $\mathcal{W}^{k-1} \equiv (\{w^{k-1}(\cdot; i, s)\}_i, V_e^{k-1}(s))_{s \in S}$ in memory & we proceed as follows:
 - substitute $w^{k-1}(\cdot)$ for $w(\cdot)$ in (1) and solve for (χ^k, x^k) for incumbents, and the entry policy.
 - That is incumbents solve for (χ^k, x^k) that

$$\max_{\chi \in \{0,1\}} \{[1 - \chi]\phi + \chi \sup_{x \geq 0} [\pi(i, s) - cx + \beta \sum_{\nu} w^{k-1}(\nu; i, s) p(\nu|x)]\}$$

- This is a Kuhn-Tucker problem for investment conditional on continuing. E.g. if ν moves up one with probability $p(x)$ or stay the same $(1 - p(x))$ then

$$\frac{\partial p(x)}{\partial x} [w(1; i, s) - w(0; i, s)] - c \leq 0,$$

with strict inequality if and only if $x = 0$.

- Substitute the solution for x in the continuation value above and determine whether it is greater than ϕ . If it is not $\chi^k = 0$ and $x^k = 0$. If it is then x^k is the solution to the Kuhn Tucker equation.
- Potential entrants.

$$\chi_e^k = 1 \Leftrightarrow \phi < V_e^k(s) \equiv \beta \sum_{\zeta} w^{k-1}(\zeta; i_e, s + e(i_e)) \mu(\zeta).$$

- Compute $w^k(\cdot|i, s)$ and $\chi_e^k(s)$, & put in memory. This is $\mathcal{W}^k$.
- Compute a distance metric between $\mathcal{W}^k$ and $\mathcal{W}^{k-1}$ (usually Euclidean in percentage change in components), and stop if sufficiently small. Else compute iteration $k + 1$.
- At fixed point each incumbent and potential entrant
 - uses, as its perceived distribution of the future states of its competitors, the actual distribution of future states of those competitors, and
 - chooses its policy to maximize its expected discounted value of future net cash flow given this distribution of the future of its competitors.

- Setting K . Start with the monopoly problem and large K ; $\Rightarrow$ a lowest i at which the monopolist remains active ($\omega = 1$) and a highest i at which the monopolist invests ($\omega = K$). Check whether at $\bar{n}$, K is large enough.
- Setting $\bar{n}$. Start at some $\bar{n}$. Then check whether $\bar{n} - 1$ and $\bar{n} + 1$ satisfy the condition of no further entry.
- When checking for $\bar{n} + 1$ use $\bar{n}$'s value function as initial condition. E.g. if $\bar{n} = 2$ then $V^0(i_1, i_2, i_3) = V^*(i_1, \max(i_2, i_3))$.
- Usually use an iterate of $\pi(i, s)$ as initial starting value (for iteration $k=0$)
- Exit can cause cycling (iteration k implies exit, then iteration $k + 1$ generates entry, ...). Use dampening procedure see problem set.
- The computed "quality ladder" model we turn to is in PM(1994). Illustrates dynamic issues & often used as a basis for checking against alternative computational techniques which I come back to below.

Computational Example.

- The exercise will start with one model similar to this, and then another where one can purchase at both a retail outlet and an internet platform.
- Table 1. Summarizes industry evolution (on recurrent class)
- Parameterization: total current cost of producing the typical output of this industry is about 25. So setting the entry cost at .2 means it is about 1/125 of the cost of production in a given period. It is not very costly to enter in our base case.
- Comparison Cases (dynamic "policy analysis"):
 - MP is base case.
 - $\sigma^* = .65$. This is an institutionally created upper bound to the max market share of the largest firm (insured by a DOJ).
 - $x^e = 2$. Increase sunk entry costs by a factor of 10. PP is planners market.
 - PP = planners' problem
 - Coll. Is "perfect" cartel (no incentive compatibility constraints).

- Base Case. Mostly 3 or 4 firms active, but lots of entry and exit which are positively correlated (as in cross-sectional data)³.

Table 1
Characteristics of Ergodic Distribution¹

$$\delta = .7 \quad \alpha = 3 \quad \beta = .925 \quad xe = .2 \quad \phi = .1$$

$$m = 5 \quad c = 5$$

No. of Time Periods	10,000				
	MP	$\sigma^* = .65$	$x^e = 2$	PP	Coll.
% with 6 firms active	.1	.2	.0	0	0
% with 5 firms active	1.6	3.1	.3	.1	.1
% with 4 firms active	35.3	33.3	1.2	5.8	1.1
% with 3 firms active	63.0	63.3	17.1	44.5	23.1
% with 2 firms active	.0	0.0	81.4	49.6	75.6
% with entry and exit	13.1	11.5	.3	10.1	10.7
% with entry only	4.8	4.8	.7	3.0	1.9
% with exit only	2.0	2.5	.6	2.3	1.7
% with entry or exit	20.1	18.7	1.6	15.4	14.2
Gross job creation	.086	.086	.031	.033	.027
Gross job destruction	.087	.088	.032	.033	.027

³Sometimes this is also true in industries over time as exit leaves an opportunity for entry), but sometimes common shocks cause negative correlation.

●Table 2:

- Lifetime Distribution very skewed.
- 1,800 firms in 10,000 periods
- Mode of lifespan distribution is 1, mean is 18.4. This is a very skewed distribution. Most entrants try something and fail. Those who do survive, persist for a very long time. They eventually exit, but they may have made a lot of money in the interim.
- Analytic reason is approximate “S” shape of sections of the value function (Myrto’s single agent lecture: Picture).
- Initially the most likely direction is downward movement, but if we get some success we increase investment until we get over the point of inflection. Once we get over the point of inflection we tend to stay over it even though investment is decreasing. This because if we start dropping down we pick up investment.

Table 2
Lifetime Distribution
(Based on 10,000 time periods)

$$\delta = .7 \quad \alpha = 3 \quad \beta = .925 \quad x_e = .2 \quad \phi = .1$$

$$m = 5 \quad c = 5$$

Life time	Percent			Implied Hazard			Cumulative Percent		
	MP	$x^e = 2$	PP	MP	$x^e = 2$	PP	MP	$x^e = 2$	PP
1	48.3	17.5	25	48.3	17.5	12.2	48.3	17.5	25.0
2	24.7	14.6	44.0	47.7	19.6	24.0	73.0	35.0	68.9
3	6.1	4.7	9.0	22.6	7.3	17.9	79.1	40.7	77.9
4	3.6	8.5	4.4	17.2	14.3	17.9	82.7	49.1	82.3
5	2.5	3.8	1.9	14.4	7.5	10.9	85.2	52.1	84.2
6	1.3	2.8	2.2	8.78	5.9	8.9	86.5	55.7	86.5
7	.8	1.9	1.4	5.9	4.5	4.1	87.3	57.2	87.9
8	.7	.9	.8	5.5	2	4.8	88.1	58.5	88.6
9	.7	.9	.5	5.9	2.9	5.7	88.8	59.5	89.2
10	.3	1.9	.7	2.7	4.6	5.7	89.1	61.3	89.9
	Number Active			Mean		Median		Standard Deviation	
MP	1,800			18.4		2		73.2	
S	103			191.5		6		428.0	
PP	1,301			19.5		2		93.9	

- Table 3:
 - Value Distribution very skewed.
 - Only 10% make positive returns, but those who make it over the initial high mortality period do very well.
 - Note the values and rates of returns of firms operating at one time will tend to be “supernormal”, but this is no indication of there being something “wrong” (some barrier to entry); it is just that those who are in tend to be those whose investments were profitable; the expected value of an entrant is just slightly more than the sunk cost of entry.

Table 3

Realized Value Distribution

$$\delta = .7 \quad \alpha = 3 \quad \beta = .925 \quad x_e = .2 \quad \phi = .1$$

$$m = 5 \quad c = 5$$

Obs/Num	Realized Values			Life Time	Sum of Realized Values		
	MP	$\sigma^* = .65$	$x^e = 2$	MP	MP	$\sigma^* = .65$	x^e
1	72.8	45.1	83.0	79	72.8	45.0	83.0
2	52.6	38.9	34.0	247	125.4	84.0	119.0
3	33.1	36.3	32.4	718	158.57	120.3	141.0
4	32.7	35.7	28.4	118	191.3	156.0	170.0
5	29.3	30.2	20.2	102	220.6	186.3	191.0
10	22.8	21.2	15.74	5	343.8	301.3	292.0
100	7.11	6.8	-4.43	215	1462.6	1313.9	292.0
150	1.81	1.1	—	37	1700.8	1508.5	—
170	.1	-.06	—	3	1717.6	1514.0	—
171	-.05	-.07	—	2	1717.5	1514.0	—
1491	-.10	-.7	—	4	1586.5	1336.1	—
1800	-4.08	—	—	15	1282.05	—	—
	Mean	Median	Std. Dev.	# Positive	Mean of Positive	# Negative	Me Neg
MP	.71	-.1	4.09	170	10.0	1630	—
$\sigma^* = .65$	.68	-.1	3.74	150	9.5	1465	—
$x^e = 2$	2.37	-1.89	11.28	329	14.2	74	—

- Table 4

- In the base case this is often a reasonably fractured industry (concentration ratio averages .39 when usually 3 or 4 firms active)
- Not shown here is that it goes through cycles of concentration that have simply to do with the logic of the technology
 - periodically a firm rises to dominate the industry and stays there for a long time.
 - then a series of negative shocks to it, or positive shocks to a competitor, occurs and it reverts to competitive. This is all internal dynamics; there is no regulatory change.
- Markups 27 to 44%. These are considered reasonable in this kind of environment. Note that they are what generates the incentives for the entrants, but the expected value of the next entrant (the one who does not enter) is less than their sunk costs.

Table 4A¹

One-Firm Concentration Ratios

	MP	$\sigma^* = .65$	$x^e = 2$	PP	Coll.
.95 quantile	.62	.61	.63	1.0	1.0
.90 quantile	.5	.5	.63	.84	1.0
.75 quantile	.43	.43	.51	.52	.72
.50 quantile	.34	.34	.5	.5	.5
.25 quantile	.33	.33	.5	.46	.5
.10 quantile	.26	.3	.34	.33	.33
.05 quantile	.25	.25	.53	.33	.33
mean	.39	.38	.51	.54	.60
standard deviation	.11	.097	.09	.20	.20

Table 4B²

Price / Cost Ratios

	MP	$\sigma^* = .65$	$x^e = 2$	Coll.
MAX	2.11	2.21	2.18	2.45
.95	1.44	1.48	1.46	2.34
.90	1.40	1.39	1.46	2.33
.75	1.33	1.33	1.40	2.33
.50	1.30	1.30	1.40	2.33
.25	1.30	1.30	1.40	2.24
.10	1.27	1.27	1.30	2.22
.05	1.27	1.27	1.30	2.20
MIN	1.24	1.23	1.27	1.29
Mean	1.33	1.32	1.41	2.30
Standard Deviation	.085	.073	.099	(.089)

Policy Analysis.

- Consumer surplus given by

$$\int \max(\omega_j - p_j + \epsilon_j) dG(\epsilon_1, \dots, \epsilon_n) \equiv \log \sum_j \exp[\omega_j - p_j]$$

- Note that there are a distribution of welfare results associated with any given set of institutions; it depends on the realizations of the random terms. We run 100 runs of 100 periods and give mean and variance & the standard error of welfare gains is $\approx 20\%$.
- With the same institutions just differences in random outcomes generate, on average, 20% standard error of outcomes. Randomness is due to the outcomes of the innovative process.

- Table 5. Initial condition always $s_0 = e(\omega_0)$. Temporary monopolies correspond to patents.
 - Difference between monopoly and MPN: consumer benefits are large in MPN and producer benefits are large in Monopoly. Monopoly also decreases total surplus by 20%.
 - Multiproduct monopolist ("perfect collusion"; no incentive compatibility constraints). Now 10% less than MPN, and both producer and consumer surplus increase. Don't want to constrain products of monopolist.
 - It would be interesting to add a platform here. The only work I know of which does this is in a two period game. Lee and Musloff's analysis of Amazon in "Two-Sided Markets Shaped By Platform-Guided Search", which is forthcoming in *Econometrica*. The game stores (App and Play) store are also platforms, and they all have different rules.

- Compare MPN to the planner's problem. Not much difference in welfare but huge difference in market structure. Planner is much more similar to colluder in observables (in entry exit rates, number of firms....). The colluder however charges much higher prices. Its P/C ratio is 2.3 whereas the MPN average is 1.3 and the planner's is one.
- Look at number of firms and investment.
 - MPN has most firms and investment. Mankiw and Whinston story; that is incumbents and potential entrants do not internalize the business stealing effects and hence "overinvest".
 - Then comes the planner; it internalizes the business stealing which causes lower investment, but it also internalizes the consumer surplus which causes higher.
 - Then the colluder; it internalizes business stealing but not consumer surplus.

- $\sigma^* = .65$. Here when an unconstrained firm would have gotten a market share greater than .65, that firm increases its price until the market share goes back to this number. This has an effect on investment (dampening it) as well as on pricing. Market structures look much the same as unconstrained MPE but welfare falls 5%.
- Increase sunk costs. Market structures change dramatically, look like colluder, but welfare does not change much (and if we added back in the sunk costs it would be even closer). Actually it is likely that there is a sunk cost in between which maximizes welfare from MPE (since when the entry costs are low there is too much entry and investment). Still there is not too much fine tuning possible here, as the MPN is pretty close to the social planner.
- The relationship between market structure and welfare is quite complex.

Table 6

Average Investment and Number of Active Firms Under Alternative Institutional Arrangements

	Investment	Active Firms
Markov Perfect Nash	2.57	3.4
$\sigma^* = .65$	2.56	3.4
$x^e = 2$	1.97	2.2
Planner	1.95	2.6
Perfect Collusion	1.75	2.3